

UML

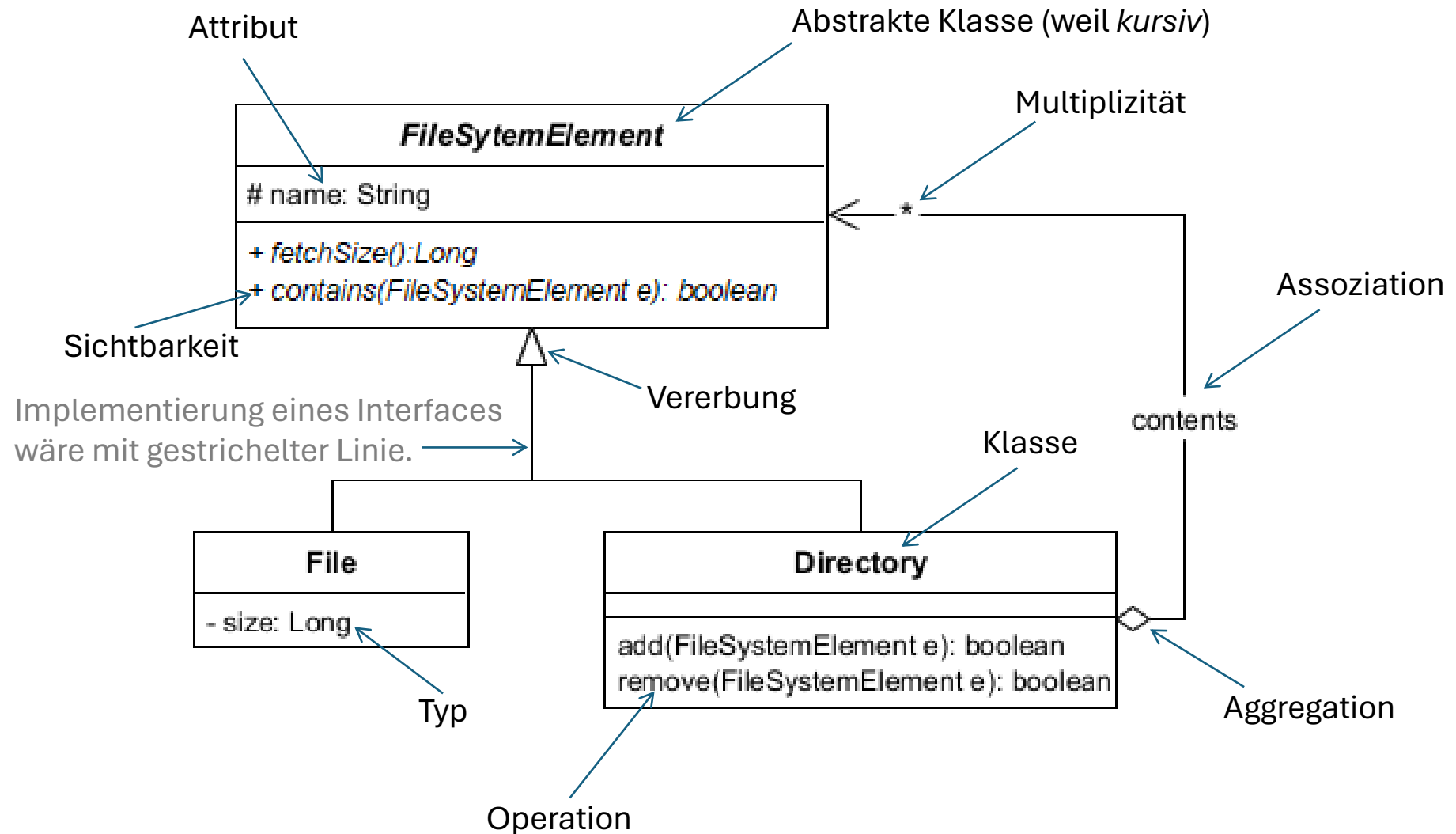
UML (Unified Modeling Language)

„The objective of UML is to provide system architects, software engineers, and software developers with tools for analysis, design, and implementation of software-based systems as well as for modeling business and similar processes.” ([UML Spezifikation 2.5.1](#))

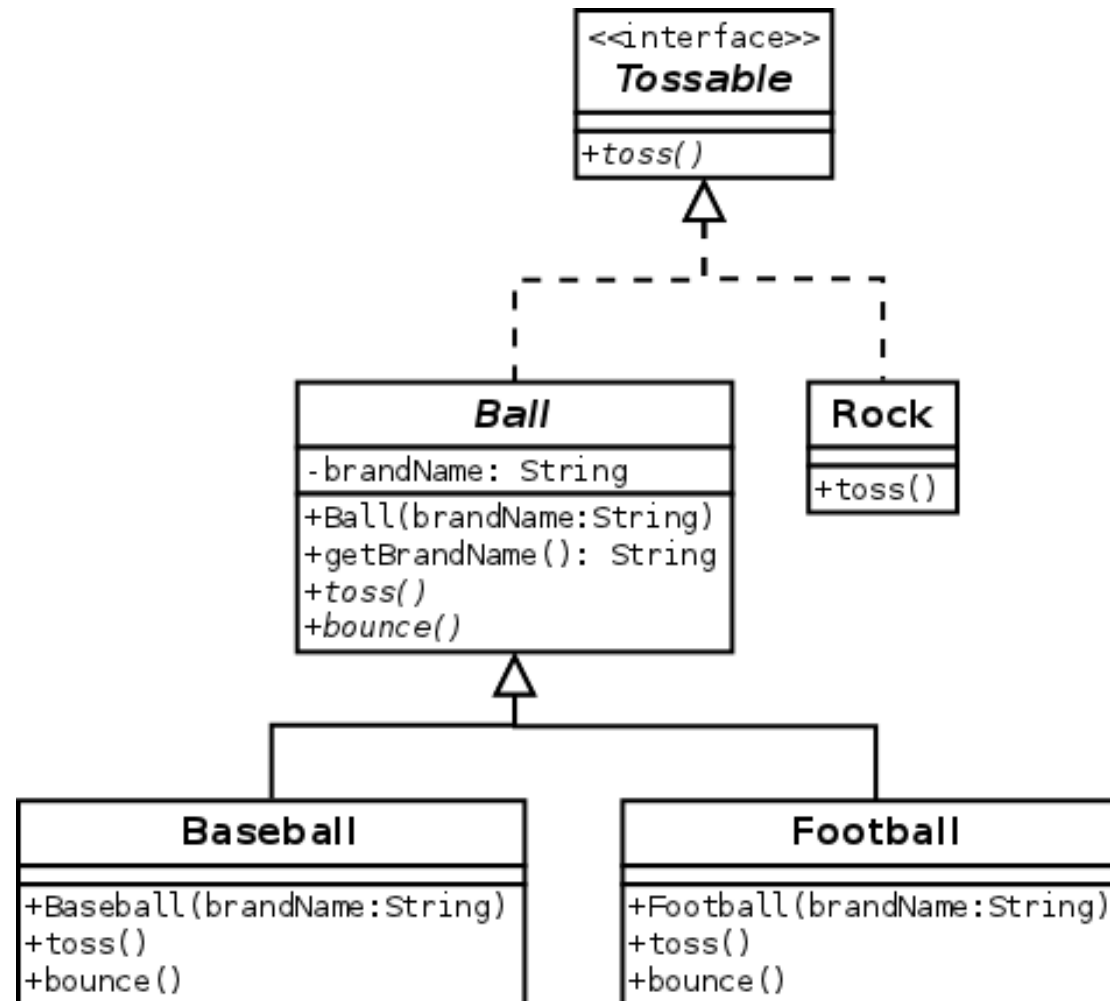
- Zur **Verständigung** zwischen Entwicklern und Fachbereichen
- Zur **Übersicht der Struktur** von **System/Prozessen/Architektur**
- Als **Dokumentation**
- **Unabhängig** von der verwendeten **Programmiersprache/Technologie**
- Für Diagramme wie Klassendiagramme lassen sich automatisiert **Code generieren**

Sieh auch: [UML-Notationsübersicht](#)

Klassendiagramme



Klassendiagramm: Abstract vs Interface



Klasse

Student

```
class Student {  
}
```

oder:

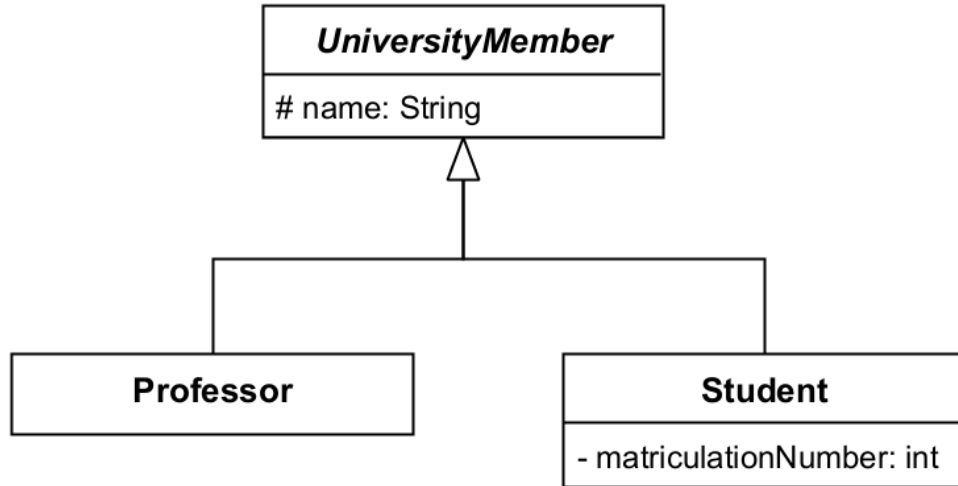
```
record Student() {  
}
```

Klasse mit Attributen

Student
- matriculationNumber: int + name: String

```
class Student {  
    private int matriculationNumber;  
    public String name; // nicht möglich mit record  
}
```

Vererbung



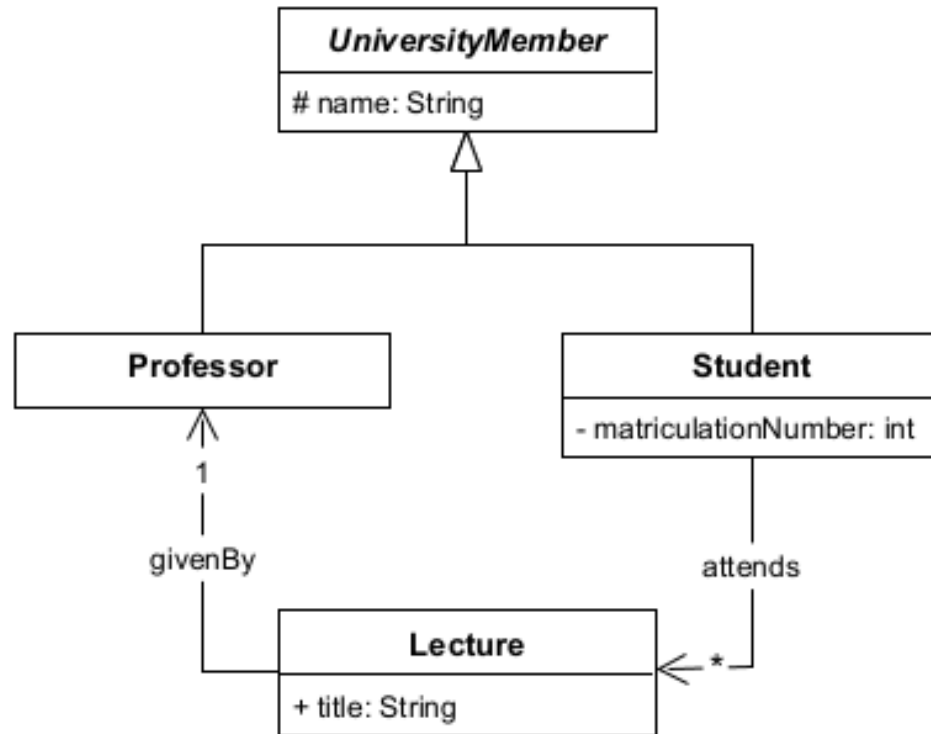
```
abstract class UniversityMember {  
    protected String name;  
}
```

```
class Professor extends UniversityMember {  
}
```

```
class Student extends UniversityMember {  
    private int matriculationNumber;  
}
```

// FYI: extends not allowed for record

Assoziationen



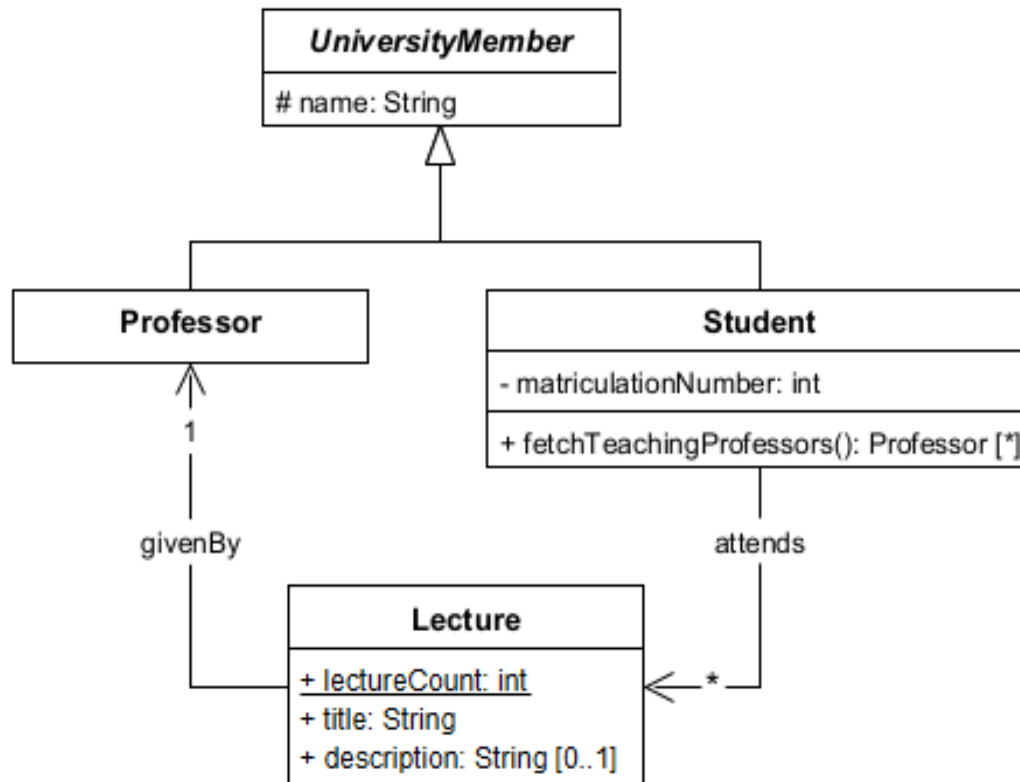
```
abstract class UniversityMember {  
    protected String name;  
}
```

```
class Professor extends UniversityMember {  
}
```

```
class Student extends UniversityMember {  
    private int matriculationNumber;  
    private Lecture[] attends;  
}
```

```
class Lecture {  
    public String title;  
    private Professor givenBy;  
}
```


Statische Attribute und Operationen



```
abstract class UniversityMember {  
    protected String name;  
}
```

```
class Professor extends UniversityMember {}
```

```
class Student extends UniversityMember {  
    private int matriculationNumber;  
    private Lecture[] attends;  
  
    public Professor[] fetchTeachingProfessors() {...}  
}
```

```
class Lecture {  
    public static int lectureCount;  
    public String title;  
    private Professor givenBy;  
}
```